Count equal groups in the sunshine

Shadows at Noon

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little draws 2 rows of chalk marks on the path, with 3 marks in each row. How many chalk marks are there?

3 + 3 = _____

Show your thinking.

2. Big sets out 3 groups of measuring pebbles, with 4 pebbles in each group. How many pebbles are there altogether?

4 + 4 + 4 = _____

Show your thinking.

3. Little lays sticks in 4 rows in the sunny yard, with 2 sticks in each row. How many sticks are there?

2 + 2 + 2 + 2 = _____

Show your thinking.

Count equal groups in the sunshine

Shadows at Noon

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Show your thinking with numbers, marks, or words.

4. A shadow notebook has 2 rows of boxes, with 5 boxes in each row. How many boxes are there?

$5 + 5 =$ _____

Show your thinking.

5. Little draws 5 groups of shadow-tip dots in the notebook, with 3 dots in each group. How many dots are there?

$3 + 3 + 3 + 3 + 3 =$ _____

Show your thinking.

6. Big draws 5 rows of chalk circles beside the shadow, with 5 circles in each row. How many circles are there?

$5 + 5 + 5 + 5 + 5 =$ _____

Show your thinking.

WONDER DAILY

Shadows at Noon

A morning question

Little Wonderer found a long shape beside a standing stick. It was a shadow: a place where the stick blocked sunlight, the light from the sun.

Save the tip

Little took a piece of chalk, a soft stick for making marks. "First mark the tip," said Big. Little drew a small line where the shadow ended.

A smaller patch

At noon, the middle of the day, they came back. The wooden stick had stayed in place, but its shadow had changed. "It is short and crisp!"

Follow the light

"Which way does it point?" asked Big. Little studied the ground. Sunlight reached one side of the stick. The shadow stretched away on the other side.

Compare the marks

Little marked the new tip with chalk. They used the same measuring stick to check both lengths.

A clue to keep

In the notebook, Little drew a long line for morning and a short line for noon. Later, when the sun was lower, the shadow would grow longer again.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Give a measurement its words

Shadows at Noon · Curiosity day

Name _____ Date _____

Write one clear command sentence about measuring a shadow. Say it first if you like.



Save the tip



Compare the marks



A clue to keep

Draw the stick, its shadow, and a chalk mark. Add one label.

stand · stick · mark · tip · shadow · measure · first · then

First, ... tell how to stand the stick or mark the tip

Then, ... tell how to measure from the stick to the mark

At the end, ... tell how to save or compare the length

Does your sentence tell what to do, with a command such as Stand, Mark, Measure, or Compare?

Shadows at Noon

01 How can we save a shadow's length?



Mark the tip, then measure from the stick to the mark.

Look closer. Little puts a chalk mark at the long shadow's tip.

A little explanation. A shadow forms where an object blocks sunlight. Mark its tip, then measure from the standing stick to the mark. Use the same measuring stick and unit to compare both lengths.

Try this. With Big, measure from the stick's base to its shadow tip. Keep your eyes on the ground; never look at the sun.

02 Why is the noon shadow so short?



Around noon, the sun is near its highest point for the day.

Look closer. The same standing stick now has a short, crisp shadow.

A little explanation. Around noon, the sun is near its highest point in the sky for the day. Light reaches the ground from higher above, so the stick's shadow stays closer to its base. A lower morning or afternoon sun makes a longer shadow.

Try this. Compare the short shadow with the earlier chalk mark.

03 Which way does a shadow point?



The bright side faces the sun. The shadow points away from the light.

Look closer. One side of the stick is bright. The shadow is on the other side.

A little explanation. The bright side faces the sun. The stick blocks some sunlight, leaving a shadow that points away from the light. Follow the shadow along the ground.

Try this. Point from the bright side, past the stick, toward the shadow.

04 What clue can a short shadow give us?



A short noon shadow is a clue that the sun is high.

Look closer. Little keeps a long line and a short line in the notebook.

A little explanation. A short noon shadow is a clue that the sun is high. Compare the same standing stick in the same place, using the same unit. Chalk marks and notes help us remember what changed. Measuring can lead to another question.

Try this. Ask one new question about tomorrow's shadow.